

GLS University

Faculty Of Computer Applications & Information Technology

Integrated MCA

Semester VIII

Capstone Project – 1 Proposal

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Project Category: Data Analytics (Using API & Visualization)

Project Title

Dog Breed Adoption Analytics (Using PetFinder API)

Introduction

Every day, thousands of dogs are listed for adoption, but not all of them find homes at the same speed. Some breeds get adopted quickly, while others wait much longer. Understanding these patterns can help shelters improve their adoption strategies.

This project uses **PetFinder API** to gather real-time data about dogs available for adoption—such as breed, age, size, and location. We analyze these details to discover which breeds get adopted fast, which ones struggle, and how adoption trends change from place to place.

Problem Statement

Shelters often don't know **why certain dogs take longer to get adopted**. Without data, it's difficult to:

- See which breeds have low adoption rates
- Understand location-based demand
- Predict how long a dog will stay in the shelter

This project aims to solve this by turning PetFinder data into clear, easy-to-understand insights.

Objectives

- Collect dog adoption data from the PetFinder API
- Analyze which breeds are adopted fast and which are slow
- Study location-wise adoption trends
- Build a clean, interactive dashboard using **React.js**
- Create a **Node.js backend** to handle data and API calls
- Present meaningful insights to support shelters and adopters

Scope of the Project

- API integration with PetFinder
- Data cleaning and processing
- Breed-wise & location-wise adoption analysis
- Visualization charts such as fastest/slowest breeds
- Interactive dashboard for users to explore trends

Significance of the Study

This project helps shelters, adopters, and organizations understand adoption behaviour better. It also gives practical experience in:

- Working with APIs
- Web development (React.js + Node.js)
- Data analysis and visualization
- Building a real-world, end-to-end project

Tools & Technologies

- **Frontend:** React.js
- **Backend:** Node.js

- **API:** PetFinder API
- **Database:** MongoDB
- **Visualization:** Chart.js / Recharts

Expected Outcome

- A working dashboard showing adoption insights
- Identification of breeds with fast or slow adoption rates
- Useful location-based adoption patterns
- A simple, clear system that helps understand how dog adoption works

Conclusion

This project brings together data, APIs, and web development to uncover real insights about dog adoption. It helps improve shelter awareness while giving hands-on experience with modern tools like React.js and Node.js.